Exercise 2

Backpropagation
SOLUTION

Please answer the following questions based on the Video we just watched together.

a) What are the three avenues you have available to nudge the activation of a neuron in the direction you want it to move?
- You can change the bias
- You can change the weights
- You can change the activations of the previous layer neurons.

b) Which weights have the strongest influence on the activation of a neuron? What "rule" to change the weights can you infer from this?
The ones associated with neurons of the previous layer with the highest activations.
Rule: Change the weights in proportion to the associated activations.

c) Similarly, to the last question, what is the rule for changing the activations?
Rule: Change the activations in proportion to the associated weights.

d) Explain the idea of stochastic gradient descent - what does it do and why? Make sure to include the concept of "mini batches" in your explanation.
Computing the gradient via backprop takes a very long time if done for all training examples. Therefore, we divide them into MiniBatches of a couple of hundred training examples and execute backprop on these. This introduces a stochastic error on the gradient, but is much faster, so we can make up for this by doing many more iterations (gradient descent steps).